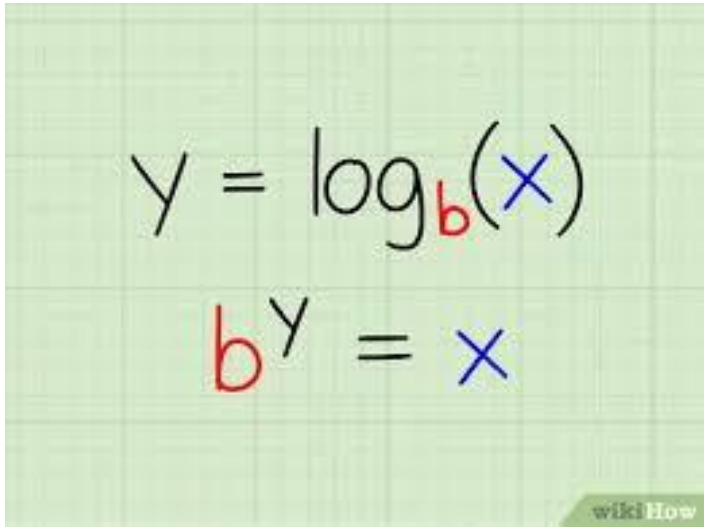
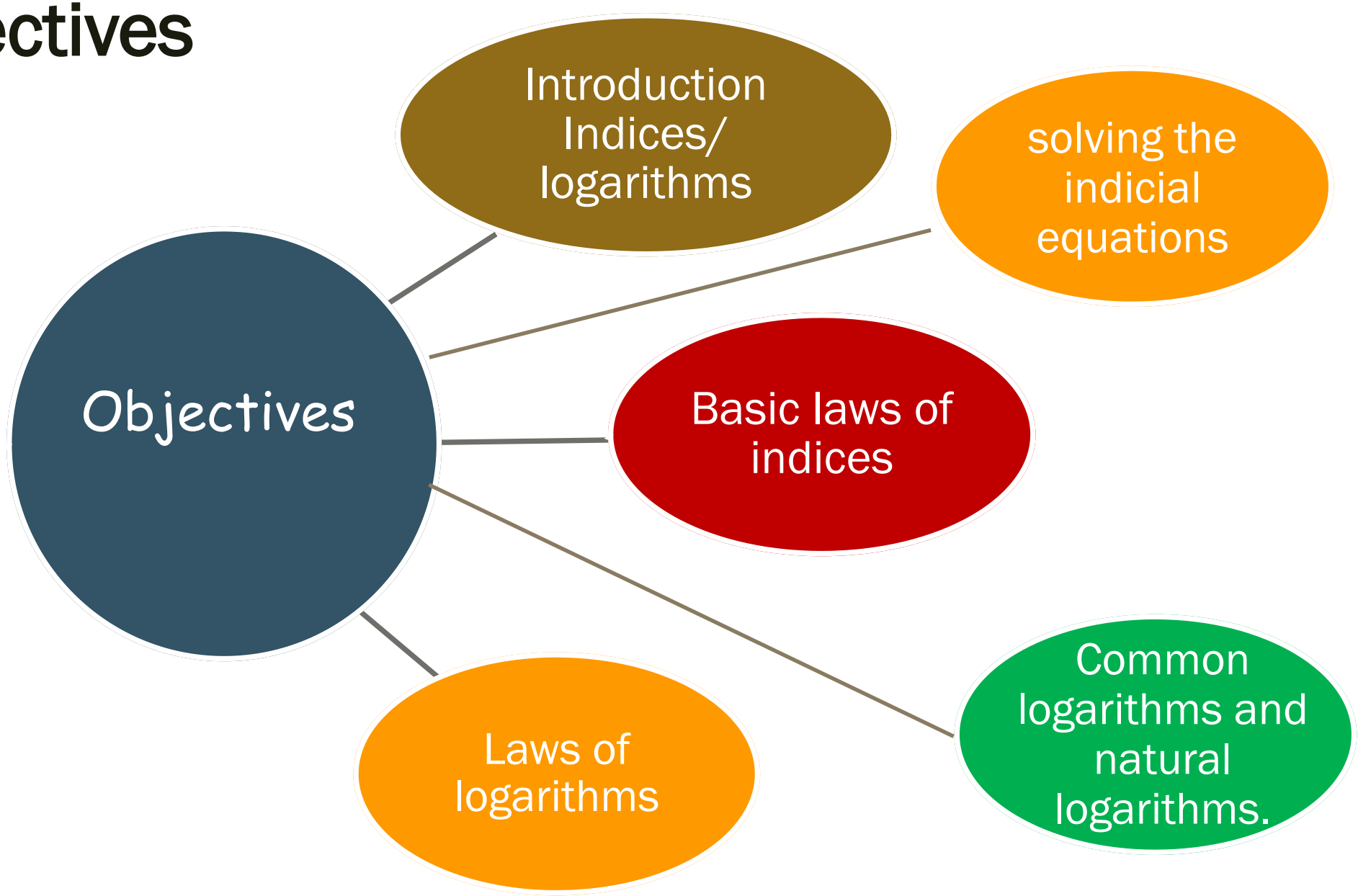
FOUNDATION OF METHEMATICS **TMS 1113**
INDICES / LOGARITHMS



Objectives


Introduction Indices/ logarithms
solving the indicial equations
Objectives
Basic laws of indices
Laws of logarithms
Common logarithms and natural logarithms.

Indices / Exponents

Indices

The number 16 is the same as $2\times2\times2\times2$, and $2\times2\times2\times2$ can be abbreviated to 2^4.
When written as 2^4, 2 is called the **base** and the 4 is called the **index** or **power**.
Sometimes we call it an **exponent**.

2^4 is read as '**two to the power of four**'.
Similarly, 3^5 is read as '**three to the power of 5**'

When the indices are 2 and 3 they are given special names; i.e. 2 is called 'squared' and 3 is called 'cubed'. Thus,
4^2 is called '**four squared**' rather than '4 to the power of 2' and

5^3 is called '**five cubed**' rather than '5 to the power of 3'

When no index is shown, the power is 1. For example, 2 means 2^1.

Evaluate (a) 2^6 (b) 3^4

(a) 2^6 means $2\times2\times2\times2\times2\times2$ (i.e. 2 multiplied by itself 6 times), and $2\times2\times2\times2\times2\times2 = \mathbf{64}$

i.e. $\mathbf{2^6 = 64}$

(b) 3^4 means $3\times3\times3\times3$ (i.e. 3 multiplied by itself 4 times), and $3\times3\times3\times3 = \mathbf{81}$

i.e. $\mathbf{3^4 = 81}$

Change the following to index form: (a) 32 (b) 625

(a) 32

(i) To express 32 in its lowest factors, 32 is initially divided by the lowest prime number, i.e. 2.

(ii) $32 \div 2 = 16$, hence $32 = 2 \times 16$.

(iii) 16 is also divisible by 2, i.e. $16 = 2 \times 8$. Thus, $32 = 2 \times 2 \times 8$.

(iv) 8 is also divisible by 2, i.e. $8 = 2 \times 4$. Thus,

$32 = 2 \times 2 \times 2 \times 4$.

(v) 4 is also divisible by 2, i.e. $4 = 2 \times 2$. Thus,

$32 = 2 \times 2 \times 2 \times 2 \times 2$.

(vi) Thus, $32 = \mathbf{2^5}$

(b) 625 .....????? $\mathbf{= 5^4}$

Square roots

When a number is multiplied by itself the product is called a square.

For example, the square of 3 is $3 \times 3 = 3^2 = 9$.

A square root is the reverse process; i.e., the value of the base which when multiplied by itself gives the number;

i.e., the square root of 9 is 3.

The symbol $\sqrt{}$ is used to denote a square root. Thus, $\sqrt{9} = 3$. Similarly, $\sqrt{4} = 2$ and $\sqrt{25} = 5$. Because $-3 \times -3 = 9$, $\sqrt{9}$ also equals -3. Thus, $\sqrt{9} = +3$ or -3 which is usually written as $\sqrt{9} = \pm 3$. Similarly, $\sqrt{16} = \pm 4$ and $\sqrt{36} = \pm 6$.

The square root of, 9 may also be written in index form as $9^{1/2}$

$$9^{\frac{1}{2}} \equiv \sqrt{9} = \pm 3$$

Practice Exercise 29 Powers and roots (answers on page 342)

Evaluate the following without the aid of a calculator.

1. 3^3
2. 2^7
3. 10^5
4. $2^4 \times 3^2 \times 2 \div 3$
5. Change 16 to index form.
6. $25^{\frac{1}{2}}$
7. $64^{\frac{1}{2}}$
8. $\dfrac{10^5}{10^3}$
9. $\dfrac{10^2 \times 10^3}{10^5}$
10. $\dfrac{2^5 \times 64^{\frac{1}{2}} \times 3^2}{\sqrt{144} \times 3}$ taking positive square roots only.

Laws of indices

There are six laws of indices.

The first law of indices
Demonstrates that **when multiplying two or more numbers having the same base, the indices are added.**

Take the example $2^2 \times 2^3$

$$
\begin{aligned}
\text{From earlier, } 2^2 \times 2^3 &= (2 \times 2) \times (2 \times 2 \times 2) \\
&= 32 \\
&= 2^5 \\
\text{Hence, } \quad 2^2 \times 2^3 &= 2^5 \\
\text{or } \quad 2^2 \times 2^3 &= 2^{2+3}
\end{aligned}
$$

The second law of indices

Demonstrates that **when dividing two numbers having the same base, the index in the denominator is subtracted from the index in the numerator.**

$$\frac{2^5}{2^3} = \frac{2\times2\times2\times2\times2}{2\times2\times2} = \frac{1\times1\times1\times2\times2}{1\times1\times1}$$

$$= \frac{2\times2}{1} = 4 = 2^2$$

Hence, $\frac{2^5}{2^3} = 2^2$ or $\frac{2^5}{2^3} = 2^{5-3}$

The third law of indices

Demonstrates that **when a number which is raised to a power is raised to a further power, the indices are multiplied.**

$$(3^5)^2 = 3^{5\times2} = 3^{10} \quad \text{and} \quad (2^2)^3 = 2^{2\times3} = 2^6$$

The fourth law of indices

States that **when a number has an index of 0, its value is 1.**

$$3^0 = 1 \quad \text{and} \quad 17^0 = 1$$

The fifth law of indices

Demonstrates that **a number raised to a negative power is the reciprocal of that number raised to a positive power.**

$$3^{-4} = \frac{1}{3^4} \quad \text{and} \quad \frac{1}{2^{-3}} = 2^3$$

Reciprocal of **5** → $\frac{1}{5}$

Reciprocal of **3/4** → $\frac{4}{3}$

The sixth law of indices,

Demonstrates that **when a number is raised to a fractional power the denominator of the fraction is the root of the number and the numerator is the power.**

$$8^{\frac{2}{3}} = \sqrt[3]{8^2} = (2)^2 = 4 \quad \text{and}$$

$$25^{\frac{1}{2}} = \sqrt[2]{25^1} = \sqrt{25^1} = \pm 5$$

(Note that $\sqrt{\ } \equiv \sqrt[2]{\ }$)

Simplify $\dfrac{a^3b^2c^4}{abc^{-2}}$

Using law (2) of indices,

$$\frac{a^3}{a} = a^{3-1} = a^2, \quad \frac{b^2}{b} = b^{2-1} = b \text{ and}$$

$$\frac{c^4}{c^{-2}} = c^{4--2} = c^6$$

Thus, $\dfrac{a^3b^2c^4}{abc^{-2}} = a^2bc^6$

Simplify $\dfrac{(mn^2)^3}{(m^{1/2}n^{1/4})^4}$

$$\frac{(mn^2)^3}{(m^{1/2}n^{1/4})^4} = \frac{m^{1\times3}n^{2\times3}}{m^{(1/2)\times4}n^{(1/4)\times4}} = \frac{m^3n^6}{m^2n^1}$$

Using law (2) of indices gives

$$\frac{m^3n^6}{m^2n^1} = m^{3-2}n^{6-1} = \boldsymbol{mn^5}$$

Simplify $\dfrac{d^2e^2f^{1/2}}{(d^{3/2}ef^{5/2})^2}$

Using law (3) of indices gives

$$\frac{d^2e^2f^{1/2}}{(d^{3/2}\,e\;f^{5/2})^2} = \frac{d^2e^2f^{1/2}}{d^3e^2f^5}$$

Using law (2) of indices gives

$$d^{2-3}e^{2-2}f^{\frac{1}{2}-5} = d^{-1}e^0f^{-\frac{9}{2}}$$

$$= d^{-1}f^{-\frac{9}{2}} \quad \text{since } e^0 = 1 \text{ from law (6) of indices}$$

$$= \frac{\mathbf{1}}{\boldsymbol{df^{9/2}}} \quad \text{from law (5) of indices}$$

Simplify $\dfrac{x^2y^3+xy^2}{xy}$

Algebraic expressions of the form $\dfrac{a+b}{c}$ can be split into $\dfrac{a}{c}+\dfrac{b}{c}$. Thus,

$$\frac{x^2y^3+xy^2}{xy}=\frac{x^2y^3}{xy}+\frac{xy^2}{xy}=x^{2-1}y^{3-1}+x^{1-1}y^{2-1}$$

$$=\boldsymbol{xy^2+y}$$

Simplify the exponential expression

$$\left(-3xy^2\right)^3$$

(a) $-27x^3y^6$

(b) $27x^3y^6$

(c) $-9xy^6$

(d) $-9x^3y^6$

Simplify the exponential expression

$$\left(\frac{-30x^5y^3}{15x^2y^{-4}}\right)^2$$

(a) $-4x^6y^{14}$

(b) $4x^6y^7$

(c) $4x^6y^{14}$

(d) $-2x^3y^7$

Perform the indicated operation and write your answer in scientific notation.

$$\frac{4,640,000,000}{232,000}$$

(a) $.2\times10^{3}$

(b) 2×10^{14}

(c) 2×10^{4}

(d) 2×10^{45}

Scientific Notation

$$a \times 10^b$$

- a is a number between 1 and 10 (called the *coefficient*),
- b is an integer (called the *exponent*), which indicates how many places the decimal point moves.
- **Large number:**
  $5,600,000 = 5.6 \times 10^6$ (Move the decimal 6 places to the left)
- **Small number:**
  $0.00042 = 4.2 \times 10^{-4}$ (Move the decimal 4 places to the right)

Example

Perform the indicated operations and write your answer in scientific notation.

$$\left(7.3\times10^{6}\right)\left(8.0\times10^{-8}\right)$$

$$\frac{4.8\times10^{7}}{3.0\times10^{-3}}$$

Example

In 2024, government proposed a 700 billion dollar bailout (7×10^{11}) of the banking system. If there are 21 million people in the country, what will be the cost per person in tax dollars if every person paid an equal amount of this bailout cost?

Practice Exercise 30 Laws of indices (answers on page 342)

Evaluate the following without the aid of a calculator.

1. $2^2 \times 2 \times 2^4$
2. $3^5 \times 3^3 \times 3$ in index form
3. $\dfrac{2^7}{2^3}$
4. $\dfrac{3^3}{3^5}$
5. 7^0
6. $\dfrac{2^3 \times 2 \times 2^6}{2^7}$
7. $\dfrac{10 \times 10^6}{10^5}$
8. $10^4 \div 10$

9. $\dfrac{10^3 \times 10^4}{10^9}$

10. $5^6 \times 5^2 \div 5^7$

11. $(7^2)^3$ in index form

12. $(3^3)^2$

13. $\dfrac{3^7 \times 3^4}{3^5}$ in index form

14. $\dfrac{(9 \times 3^2)^3}{(3 \times 27)^2}$ in index form

15. $\dfrac{(16 \times 4)^2}{(2 \times 8)^3}$

16. $\dfrac{5^{-2}}{5^{-4}}$

17. $\dfrac{3^2 \times 3^{-4}}{3^3}$

18. $\dfrac{7^2 \times 7^{-3}}{7 \times 7^{-4}}$

19. $\dfrac{2^3 \times 2^{-4} \times 2^5}{2 \times 2^{-2} \times 2^6}$

20. $\dfrac{5^{-7} \times 5^2}{5^{-8} \times 5^3}$

Example

Simplify:

$(7x^5)(-9x^3)$

$\frac{48x^5y^3z^2}{16x^2y}$

-5^{-2}

$(b^5)^4$

$(3x^4)^3$

$(5x^3y)(4x^5y^2)$

$\frac{6^5z^0}{6^5}$

$3x^{-2}y^4$

$(-3^2)^2$

$\left(\frac{x^4}{y^3}\right)^2$

$\frac{(-2)^7}{(-2)^3}$

-4^4

$(y^4)^5$

$\frac{8x^4y3}{-2x}$

x^7x^{-7}

$\frac{1}{7^{-2}}$

$(6x^3y)^2$

$\left(\frac{5x^{-2}}{y^3}\right)^3$

$(-5)^{-2}$

LOGARITHMS (Logs)

Where Did Logs Come From?

- The invention of logs in the early 1600s fueled the scientific revolution. Back then scientists, astronomers especially, used to spend huge amounts of time crunching numbers on paper.

- By cutting the time they spent doing arithmetic, logarithms effectively gave them a longer productive life.

- Logarithms were originally developed to simplify complex arithmetic calculations. They were designed to transform multiplicative processes into additive ones.

If at first this seems like no big deal,
then try multiplying
2,234,459,912 and 3,456,234,459.

Without a calculator !

Clearly, it is a lot easier to add these two numbers.

Today of course we have calculators and scientific notation to deal with such large numbers.

So at first glance, it would seem that logarithms have become obsolete.

Our first question then must be:

What is a logarithm ?

Of course logarithms have a precise mathematical definition just like all terms in mathematics. So let's start with that.

Definition of Logarithm

Suppose b>0 and b≠1,
there is a number ‘p’
such that:

$$\log_b n = p \quad if \quad and \quad only \quad if \quad b^p = n$$

We read this as: ”the log base b of n is equal to p”.

In general, if a number *y* can be written in the form a^x , then the index *x* is called the ‘logarithm of *y* to the base of *a*’, i.e.

$$\text{if } y = a^x \text{ then } x = \log_a y$$

There are still good reasons for studying them.

- To model many natural processes, particularly in living systems. We perceive loudness of sound as the logarithm of the actual sound intensity, and dB (decibels) are a logarithmic scale.
- To measure the pH or acidity of a chemical solution.
- To measure earthquake intensity on the Richter scale.

How they are developed...?

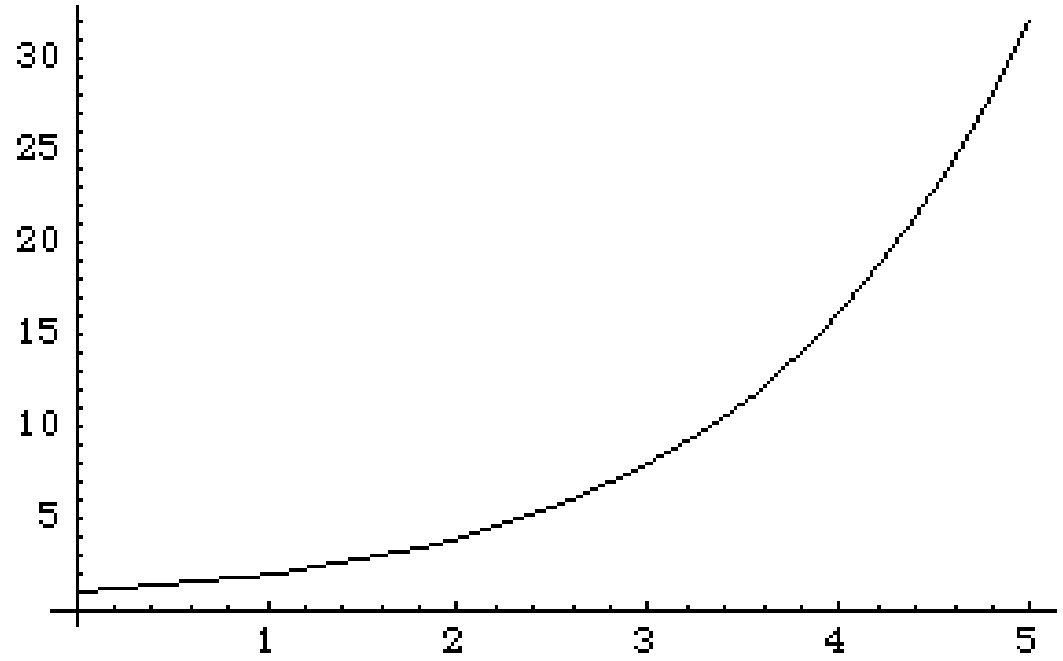
- In the mathematical operation of addition we take two numbers and join them to create a third $4 + 4 = 8$
- We can repeat this operation: $4 + 4 + 4 = 12$
- Multiplication is the mathematical operation that extends this: $3 \bullet 4 = 12$
- In the same way, we can repeat multiplication: $3 \bullet 3 \bullet 3 = 27$
- The extension of multiplication is exponentiation: $3 \bullet 3 \bullet 3 = 27 = 3^3$

Now consider that we have a number and we want to know how many 2's must be multiplied together to get that number.

- For example, given that we are using `2' as the base, how many 2's must be multiplied together to get 32? That is, we want to solve this equation: $2^B = 32$
- Of course, $2^5 = 32$, so $B = 5$. To be able to get a hold of this, mathematicians made up a new function called the logarithm: $\log_2 32 = 5$

More on development

- The exponential function $y = 2^x$ is shown in this graph:



Logarithms

Common logarithms

From the above, if we write down that $1000 = 10^3$, then $3 = \log_{10} 1000$. This may be checked using the 'log' button on your calculator.

Logarithms having a base of 10 are called **common logarithms** and $\log_{10}$ is usually abbreviated to lg. The following values may be checked using a calculator.

$$\lg 27.5 = 1.4393\ldots$$

$$\lg 378.1 = 2.5776\ldots$$

$$\lg 0.0204 = -1.6903\ldots$$

Natural logarithms

Logarithms having a base of e (where e is a mathematical constant approximately equal to 2.7183) are called **hyperbolic, Napierian** or **natural logarithms**, and $\log_e$ is usually abbreviated to ln. The following values may be checked using a calculator.

$$\ln 3.65 = 1.2947\ldots$$

$$\ln 417.3 = 6.0338\ldots$$

$$\ln 0.182 = -1.7037\ldots$$

Logarithms having a base of 10 are called **common logarithms** and log10 is usually abbreviated to lg

$$\lg 27.5 = 1.4393\ldots$$
$$\lg 378.1 = 2.5776\ldots$$
$$\lg 0.0204 = -1.6903\ldots$$

Logarithms having a base of *e* (where *e* is a mathematical constant approximately equal to 2.7183) are called **hyperbolic, Napierian** or **natural logarithms**, and $\log_e$ is usually abbreviated to ln.

$$\ln 3.65 = 1.2947\ldots$$
$$\ln 417.3 = 6.0338\ldots$$
$$\ln 0.182 = -1.7037\ldots$$

Example 1:

Write $2^3 = 8$ in logarithmic form.

Solution: $\log_2 8 = 3$

We read this as: ”the log base 2 of 8 is equal to 3”.

Example 2:

$$Write \;\; 2^{-3} = \frac{1}{8} \;\; in \;\; \log arithmic \;\; form.$$

Solution: $\log_2 \frac{1}{8} = -3$

Read as: "the log base 2 of $\frac{1}{8}$ is equal to -3".

Exercises

1. *Write* $7^2 = 49$ *in* $\log$*arithmic form.*

Solution: $\log_7 49 = 2$

2. *Write* $5^0 = 1$ *in* $\log$*arithmic form.*

Solution: $\log_5 1 = 0$

3. *Write* $10^{-2} = \frac{1}{100}$ *in* $\log$*arithmic form.*

Solution: $\log_{10} \frac{1}{100} = -2$

Here are some worked problems to help understanding of logarithms.

Problem 1. Evaluate $\log_3 9$

Let $x = \log_3 9$ then $3^x = 9$ from the definition of a logarithm,

i.e. $3^x = 3^2$, from which $x = 2$

Hence, $\mathbf{\log_3 9 = 2}$

Problem 2. Evaluate $\log_{10} 10$

Problem 3. Evaluate $\log_{16} 8$

Problem 4. Evaluate $\lg 0.001$

Let $x = \lg 0.001 = \log_{10} 0.001$ then $10^x = 0.001$

i.e. $10^x = 10^{-3}$

from which, $x = -3$

Hence, $\mathbf{\lg 0.001 = -3}$ (which may be checked using a calculator)

Problem 5. Evaluate $\ln e$

Let $x = \ln e = \log_e e$ then $e^x = e$

i.e. $e^x = e^1$, from which

$x = 1$

Hence, $\mathbf{\ln e = 1}$ (which may be checked by a calculator)

Problem 6. Evaluate $\log_3 \frac{1}{81}$

Problem 7. Solve the equation $\lg x = 3$

Problem 8. Solve the equation $\log_2 x = 5$

If $\log_2 x = 5$ then $\boldsymbol{x = 2^5 = 32}$

Problem 9. Solve the equation $\log_5 x = -2$

If $\log_5 x = -2$ then $x = 5^{-2} = \frac{1}{5^2} = \mathbf{\frac{1}{25}}$

Logarithm Rules

These rules are important. You need to master them before you try to solve logarithmic equations. Apply them to the problems on the next 5 slides to see if you understand.

- $\ln A \cdot B = \ln A + \ln B$
- $\log_a x \cdot y = \log_a x + \log_a y$
- $\ln x/y = \ln x - \ln y$
- $\log_a x/y = \log_a x - \log_a y$
- $\ln x^b = b \cdot \ln x$
- $\log_a x^b = b \cdot \log_a x$

Some important instructions to remember

$Log_a 1 = 0$ $a^0 = 1$

$a^1 = a$

$$\log_a a = 1$$

$$\log_a \sqrt[n]{x} = \log_a x^{1/n} = \frac{1}{n}\log_a x$$

$$\log_a\left(\frac{1}{x}\right) = \log_a x^{-1} = -\log_a x$$

Problem 10. Write $\log 4 + \log 7$ as the logarithm of a single number

$\log 4 + \log 7 = \log(7 \times 4)$ by the first law of logarithms

$= \mathbf{\log 28}$

Problem 14. Simplify $\log 64 - \log 128 + \log 32$

$64 = 2^6$, $128 = 2^7$ and $32 = 2^5$

Hence, $\log 64 - \log 128 + \log 32$

$= \log 2^6 - \log 2^7 + \log 2^5$

$= 6\log 2 - 7\log 2 + 5\log 2$

by the third law of logarithms

$= \mathbf{4\log 2}$

Problem 15. Write $\frac{1}{2}\log 16 + \frac{1}{3}\log 27 - 2\log 5$ as the logarithm of a single number

$$\frac{1}{2}\log 16 + \frac{1}{3}\log 27 - 2\log 5$$

$$= \log 16^{\frac{1}{2}} + \log 27^{\frac{1}{3}} - \log 5^2$$

by the third law of logarithms

$$= \log \sqrt{16} + \log \sqrt[3]{27} - \log 25$$

by the laws of indices

$$= \log 4 + \log 3 - \log 25$$

$$= \log\left(\frac{4 \times 3}{25}\right)$$ by the first and second laws of logarithms

$$= \log\left(\frac{12}{25}\right) = \mathbf{\log 0.48}$$

Problem 16. Write (a) log 30 (b) log 450 in terms of log 2, log 3 and log 5 to any base

Problem 20. Solve the equation $\frac{1}{2}\log 4 = \log x$

Problem 21. Solve the equation

$$\log\left(x^2-3\right)-\log x=\log 2$$

Indicial equations

The laws of logarithms may be used to solve certain equations **involving powers**, called **indicial equations**.

For example, to solve, say, $3^x = 27$, logarithms to a base of 10 are taken of both sides,

i.e. $\log_{10} 3^x = \log_{10} 27$

and $x \log_{10} 3 = \log_{10} 27$

by the third law of logarithms

Rearranging gives $x = \dfrac{\log_{10} 27}{\log_{10} 3} = \dfrac{1.43136\ldots}{0.47712\ldots}$

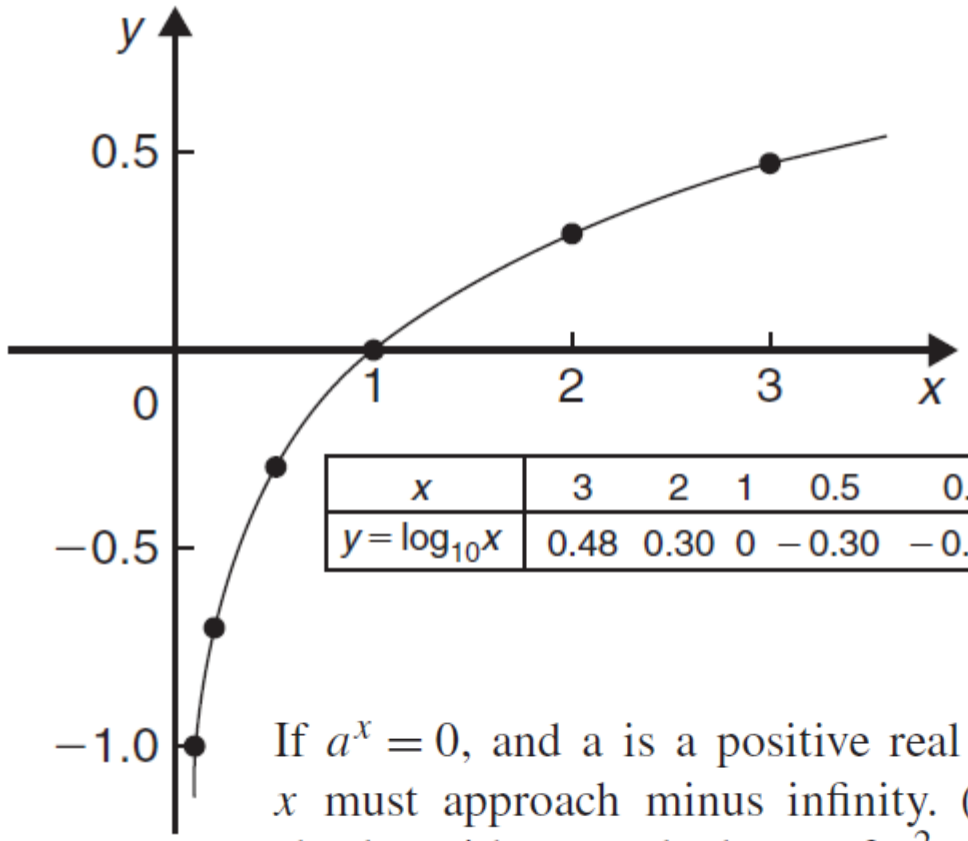
$= \mathbf{3}$ which may be readily checked.

$\left(\text{Note, } \dfrac{\log 27}{\log 3} \text{ is } \mathbf{not} \text{ equal to } \log \dfrac{27}{3}\right)$

Problem 23. Solve the equation $2^{x+1} = 3^{2x-5}$ correct to 2 decimal places

Graphs of logarithmic functions

A graph of $y = \log_{10} x$ is shown in following figure.



x	3	2	1	0.5	0.2	0.1
$y = \log_{10} x$	0.48	0.30	0	−0.30	−0.70	−1.0

If $a^x = 0$, and a is a positive real number, then x must approach minus infinity. (For example, check with a calculator, $2^{-2} = 0.25, 2^{-20} = 9.54 \times 10^{-7}, 2^{-200} = 6.22 \times 10^{-61}$, and so on.)
Hence, $\mathbf{\log_a 0 \to -\infty}$

In general, with a logarithm to any base, *a*, it is noted that

(a) $\mathbf{\log_a 1 = 0}$

Let $\log_a = x$ then $a^x = 1$ from the definition of the logarithm.

If $a^x = 1$ then $x = 0$ from the laws of logarithms.

Hence, $\mathbf{\log_a 1 = 0}$. In the above graphs it is seen that $\log_{10} 1 = 0$ and $\log_e 1 = 0$.

(b) $\mathbf{\log_a a = 1}$

Let $\log_a a = x$ then $a^x = a$ from the definition of a logarithm.

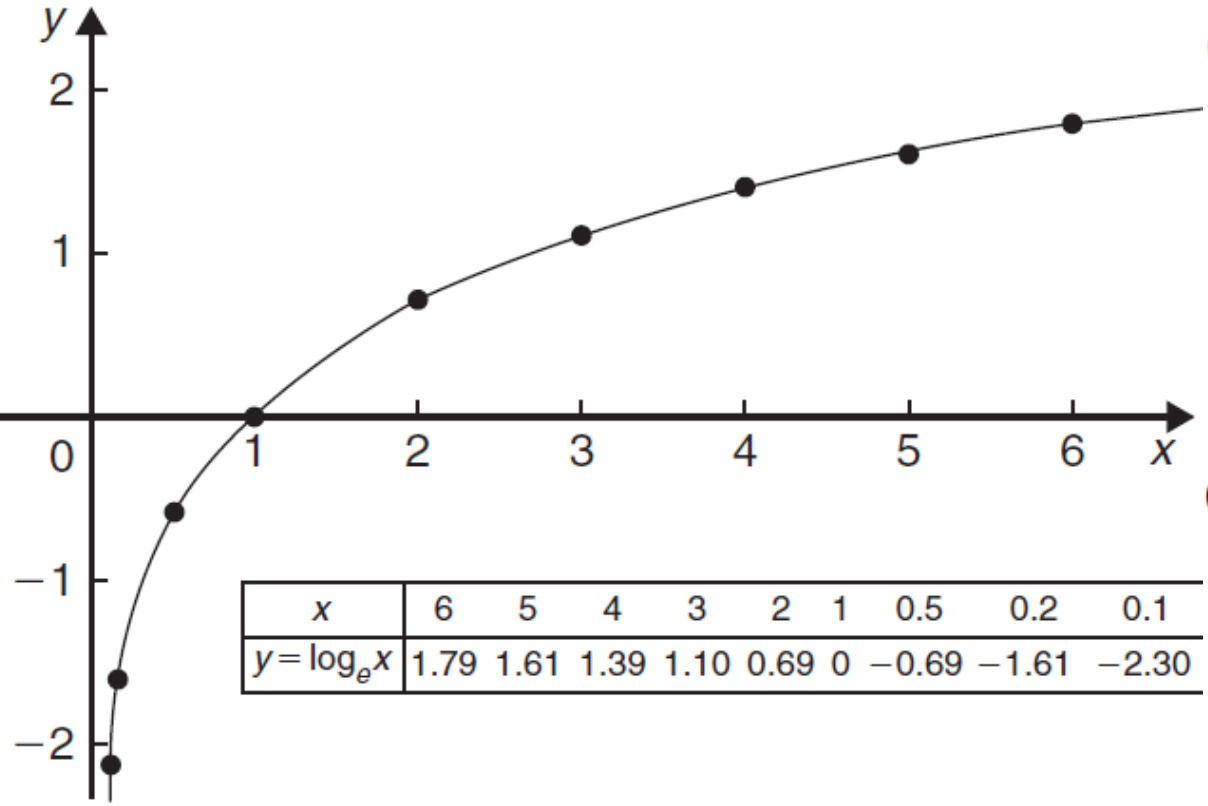
If $a^x = a$ then $x = 1$.

Hence, $\mathbf{\log_a a = 1}$. (Check with a calculator that $\log_{10} 10 = 1$ and $\log_e e = 1$.)

$\mathbf{\log_a 0 \to -\infty}$

Let $\log_a 0 = x$ then $a^x = 0$ from the definition of a logarithm.

A graph of $y = \log_e x$ is shown in in following figure.



x	6	5	4	3	2	1	0.5	0.2	0.1
$y = \log_e x$	1.79	1.61	1.39	1.10	0.69	0	−0.69	−1.61	−2.30

If $a^x = 0$, and a is a positive real number, then x must approach minus infinity. (For example, check with a calculator, $2^{-2} = 0.25, 2^{-20} = 9.54 \times 10^{-7}, 2^{-200} = 6.22 \times 10^{-61}$, and so on.)
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In general, with a logarithm to any base, *a*, it is noted that

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Hence, $\mathbf{\log_a 1 = 0}$. In the above graphs it is seen that $\log_{10} 1 = 0$ and $\log_e 1 = 0$.

(b) $\mathbf{\log_a a = 1}$

Let $\log_a a = x$ then $a^x = a$ from the definition of a logarithm.

If $a^x = a$ then $x = 1$.

Hence, $\mathbf{\log_a a = 1}$. (Check with a calculator that $\log_{10} 10 = 1$ and $\log_e e = 1$.)

$\mathbf{\log_a 0 \to -\infty}$

Let $\log_a 0 = x$ then $a^x = 0$ from the definition of a logarithm.

Homework

Practice Exercise 31 Further problems on indices (answers on page 342)

Practice Exercise 59 Laws of logarithms (answers on page 346)

Practice Exercise 60 Laws of logarithms (answers on page 346)

Practice Exercise 61 Indicial equations (answers on page 346)

End of Lesson